

1 bis. HCI Overview

Interazione Uomo Macchina
Prof. Maurizio Mancini



DIPARTIMENTO
DI INFORMATICA
SAPIENZA
UNIVERSITÀ DI ROMA

Human Computer Interaction
Copyright 2023 Luigi De Russis
<https://github.com/polito-hci-2022>





Expectations?

What do you hope to learn in this course?

Goals

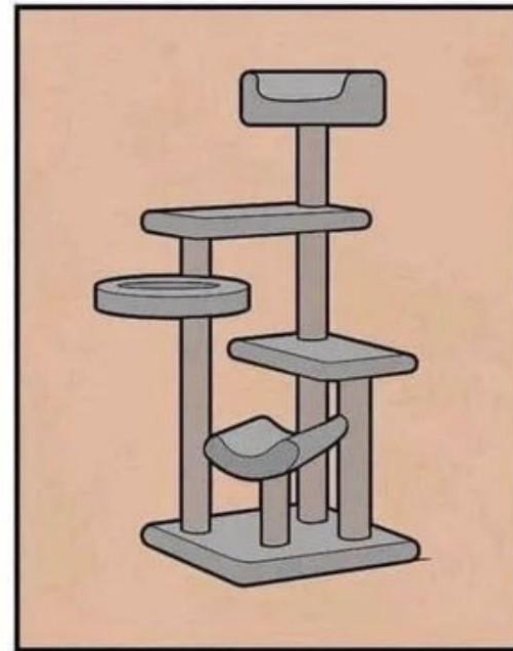
- Understanding how to design the user experience when interacting with modern applications, devices, and environments
- Gaining in-depth knowledge of a human-centered process to create interactive systems
 - and how to apply it in practice
- Becoming familiar with methods to gather and listen to users' needs
- Learning to evaluate interactive systems with their users

Why?

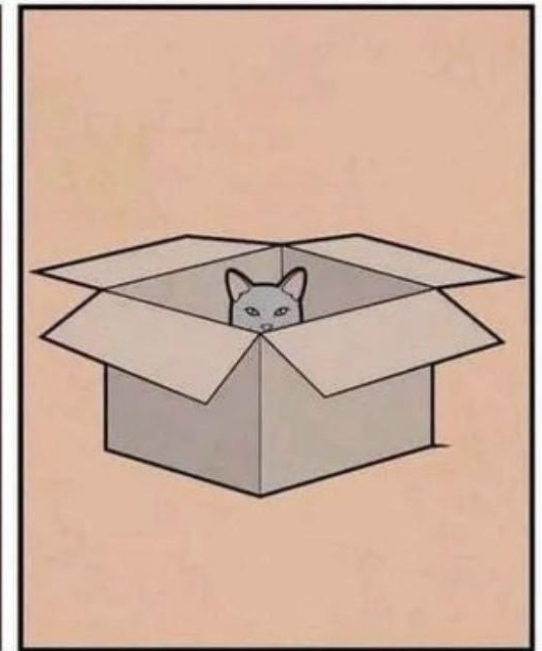


source: https://www.instagram.com/p/CT8qVYaDE_R/

**Product
features**

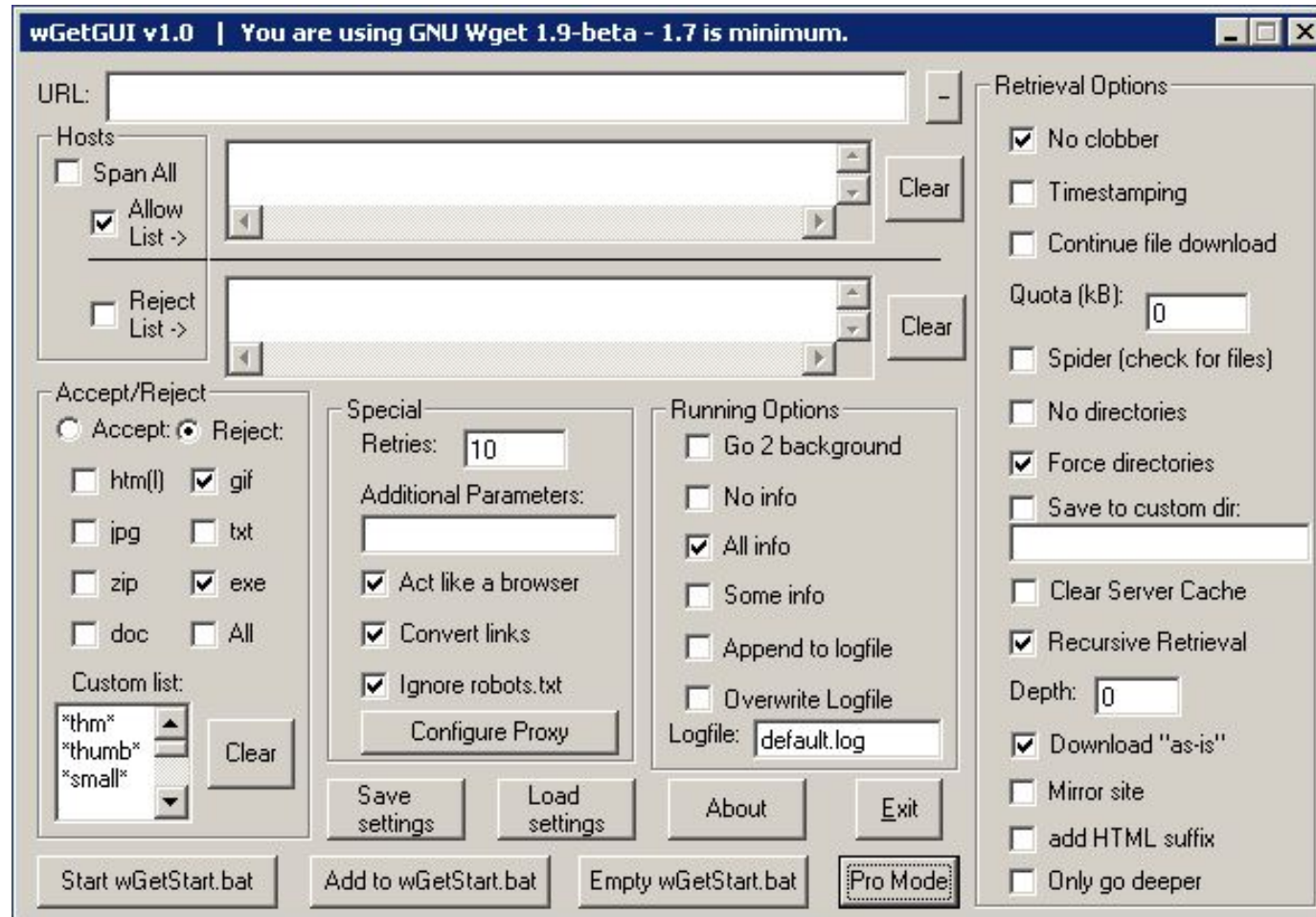


**User
needs**



© _yes_but

Hall of Fame or Shame?



How to Design and Develop Good Interactive Systems?

- *Iterative and human-centered process*
- People needs (not “wants”)
- Design principles and guidelines
- Usability goals
- Prototyping (rapidly)
- Evaluation (various kind)
- Programming

What We Will Learn

Introduction to Human-Computer Interaction	Definitions, the human, the computer, vision of the future
	Building interactive applications with a human-centered process
	Main tasks and methods to design, develop, and evaluate an interactive application
	Needfinding strategies, low-, medium, and high-fidelity prototypes, mental models and visual design, heuristic evaluation, and basic concepts and methods for user studies
	Application & Projects
	Practical part on a specific application domain
	Various kind of interactive prototypes (with and without writing code)
	“Beyond WIMP” paradigms
	Tangible interaction, wearables, voice user interfaces, gestures, interaction with AI systems, ...
	Thematic seminars on emerging topics and case studies

Course Topics... At a Glance!

1. Introduction
2. Needfinding
3. Defining needs and tasks
4. Paper Prototyping
5. Heuristic evaluation
6. Usability testing
7. Design guidelines, principles, and heuristics



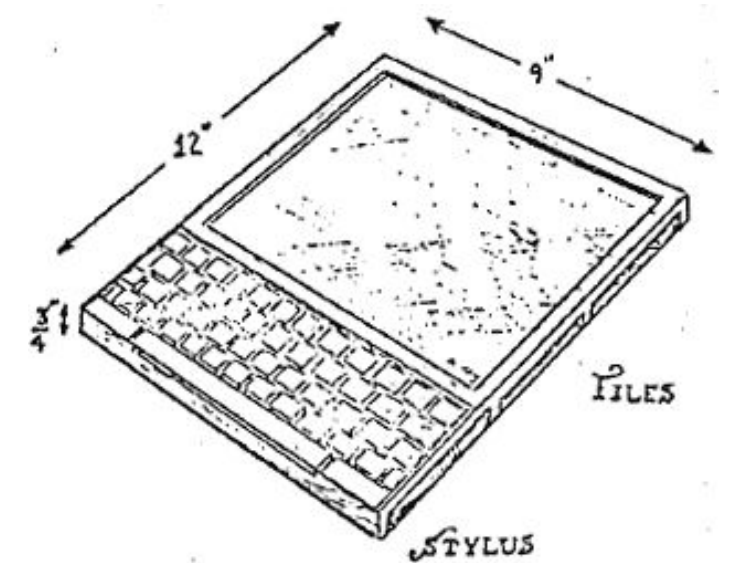
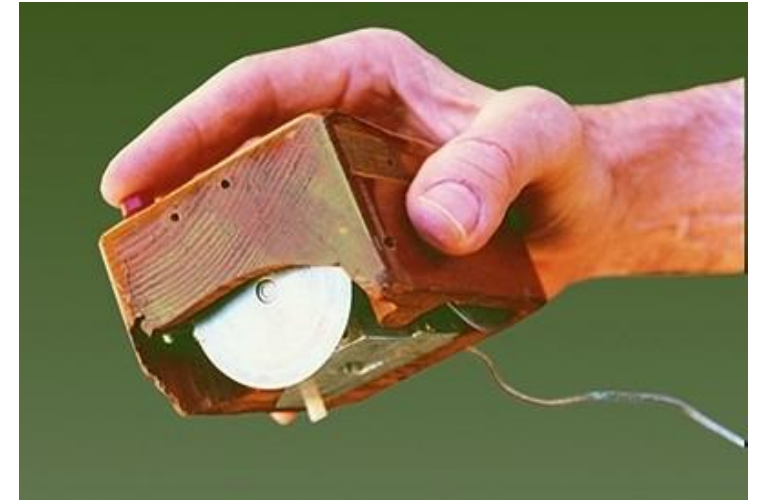
What is HCl?

Goals

- What is HCI?
- What is usability?
- What are the Interaction Design processes, and how does they relate with Software Engineering processes?
- What is meant by User Centered Design?

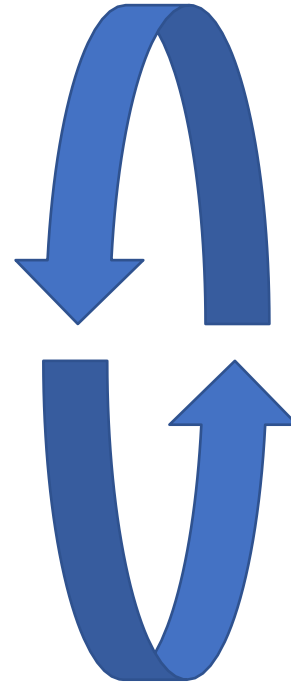
Human-Computer Interaction... In Brief

- A multi-disciplinary field
- Concerned with the **design, evaluation, and implementation** of interactive computing systems for human use
 - and with the study of **major phenomena** surrounding them
- Involves two *entities* (the human and the computer) that determine each other behavior over time
 - framed in terms of humans' **goals** and related **tasks/pursuits**



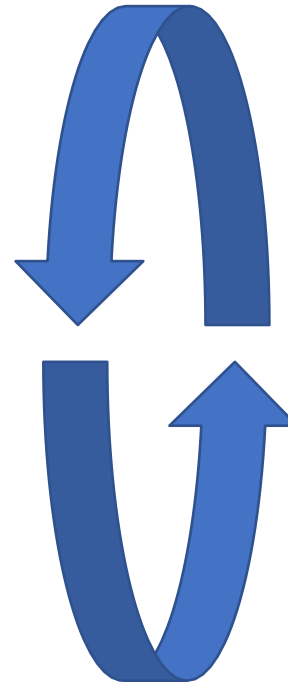
HCI Is Multidisciplinary

- Psychology and cognitive science
 - User perceptual, cognitive and problem-solving skills
- Ergonomics
 - User's physical capabilities
- Sociology
 - Understanding the wider context of the interaction
- **Computer Science and Computer Engineering**
 - **Building the necessary artifacts (HW, SW)**
- Business
 - Satisfying market needs
- Graphic design
 - Produce an effective interface presentation
- Technical writing
 - Documentation, manuals, on-screen content
- ...



HCI Is Multidisciplinary

- Psychology and cognitive science
 - User perceptual, cognitive and problem-solving skills
- Ergonomics
 - User's physical capabilities
- Sociology
 - Understanding the wider context of the interaction
- **Computer Science and Computer Engineering**
 - **Building the necessary artifacts (HW, SW)**
- Business
 - Satisfying market needs
- Graphic design
 - Produce an effective interface presentation
- Technical writing
 - Documentation, manuals, on-screen content
- ...



To help us in applying expertise from many different fields:

- Design methods and processes
- Models
- Heuristics
- Best practices
- Conventions
- Experiments and user studies

The Goal of HCI

Ingredients

- The **User(s)**
- The **Computer(s)**
- The **Task(s)** to be accomplished

Goal

- The system must support the user's **task**, with a focus on its **usability**
 - Useful
 - Usable
 - Used

The Ingredients

The human

- Sensory systems
 - Visual
 - Auditory
 - Haptic
 - Spatial
- Acting systems
 - Hands
 - Voice
 - Head, Body, ...
- Cognitive processes
 - Perception
 - Memory

The computer

- Input peripherals
 - Keyboard, mouse
 - Trackpad, trackball
 - Touch surfaces or screens
 - Microphone
 - Sensors
 - Card readers
 - ...
- Output peripherals
 - Screen
 - Audio (voice, sounds)
 - Haptics
 - VR/AR headsets
 - ...

Models of Interaction

A general framework to understand how User and System interact

What Is Interaction (in HCI)?

- Interaction...
 - **is not** the idea promoted and repeated in folk notions that a computer and a human are engaged
 - it concerns two entities – humans and computers – that determine each other's behavior over time
 - Their mutual determination can be of many types, including statistical, mechanical, and structural
- **Users**, with their **goals** and **pursuits**, are the ultimate metric of interaction

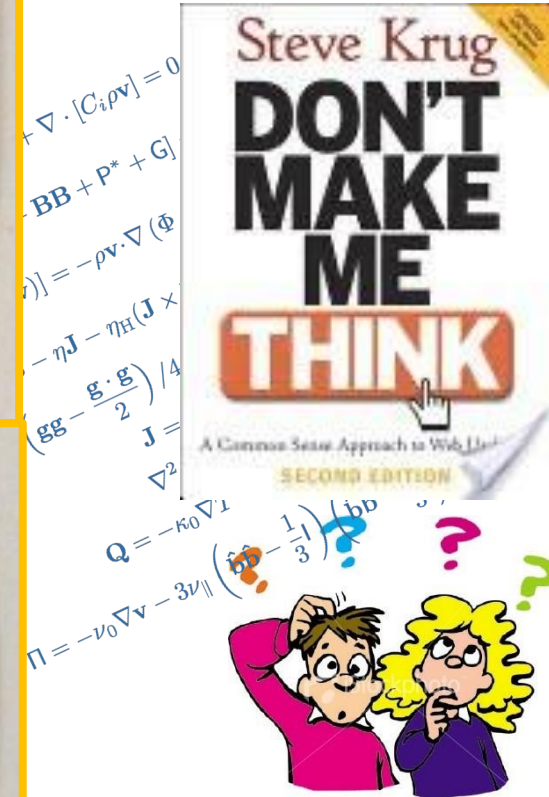
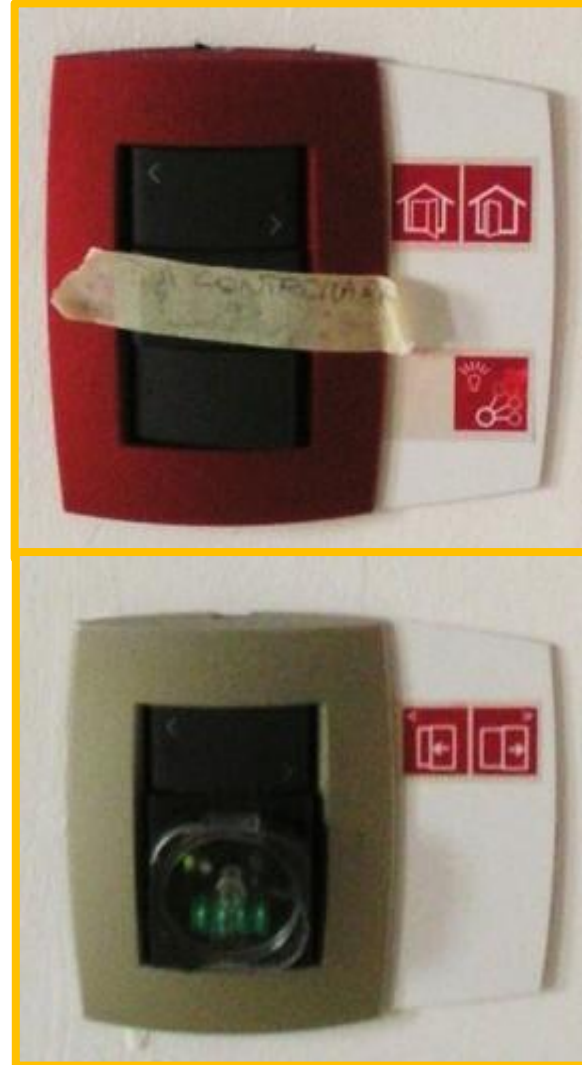
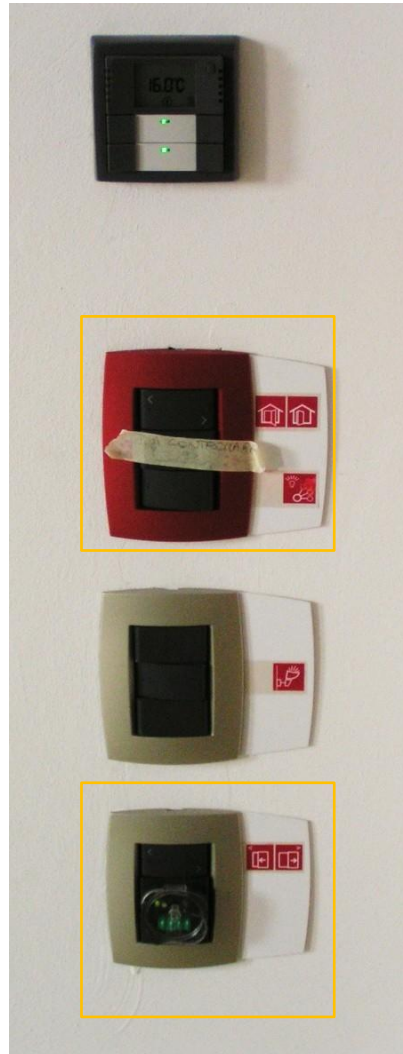
Assumptions

- The **user** wants to accomplish some **goals**, in a specific application **domain**
 - Each domain has a specific jargon, set of possible processes and goals, artifacts and building blocks, ...
- **Tasks** are operations to manipulate the concepts of a domain
 - The goal is attained by performing one or more tasks
- Interaction studies the relation between User and System
 - The system possesses a **state** and “speaks” a **core language**
 - The user possesses a **state**, that includes an **understanding** of the system’s state, some **intention** to perform a task, and “speaks” the **task language**

* About Human Errors

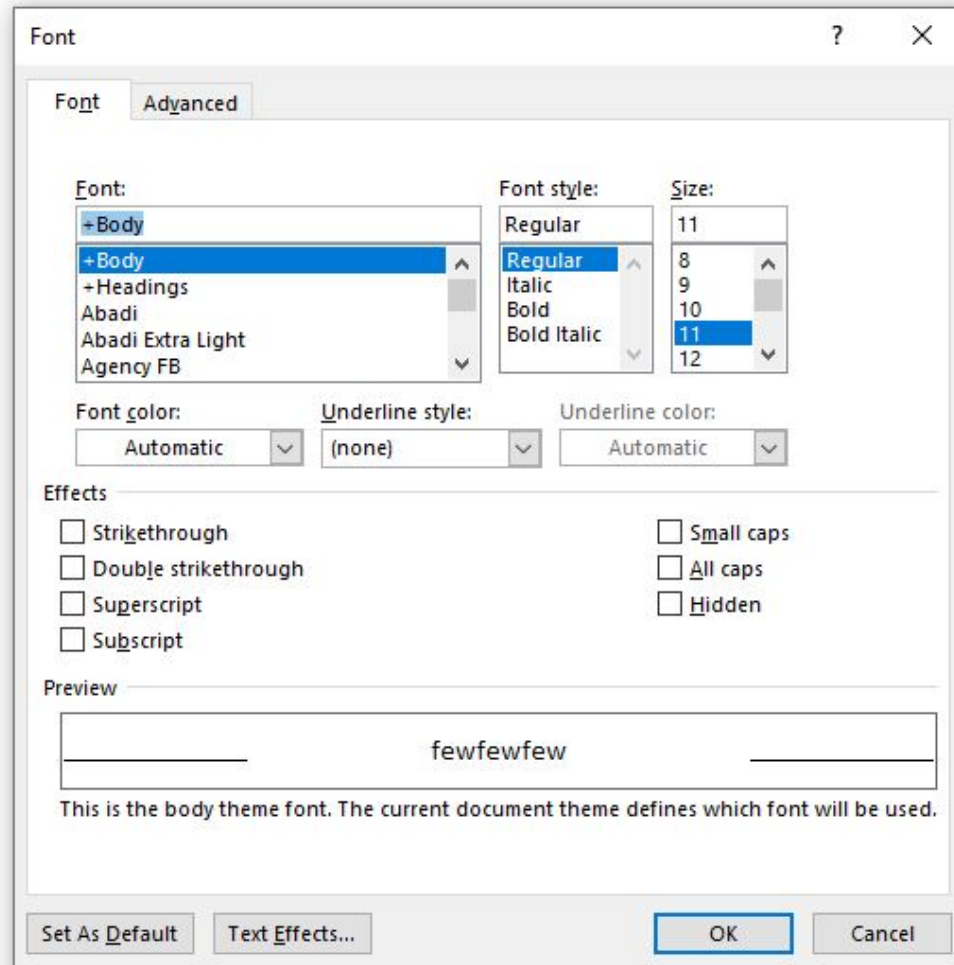
- Human errors should **never** be considered as faults of the user
- Rather, «they are usually a result of bad design» (Norman)
- Humans tend to be imprecise, distracted, not-omniscient
 - System design should anticipate this human behavior
 - Minimize the chance of inappropriate actions (evaluation)
 - Maximize the possibility of discovering and repairing an inappropriate action (execution)
 - Enable users to understand the state of the system and build an appropriate model

Example (articulation): find the right switch



Example (presentation):

Which are the allowed combinations?



Design Processes and Frameworks

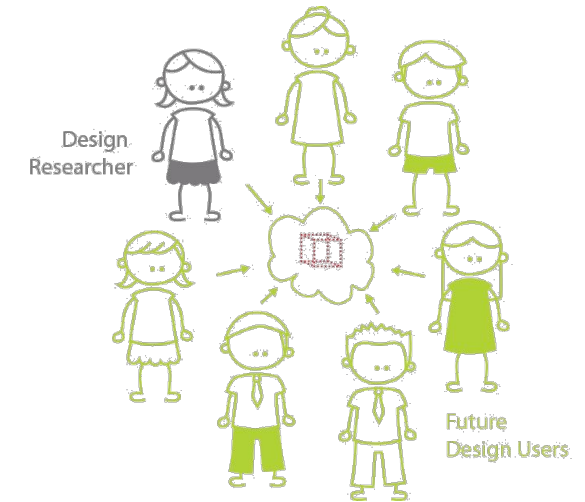
Approaches for shaping the design process

User-Centered Design (UCD)

- Avoid the risk of software project failure
 - Estimated 50% are affected by bad developer<->user/client communication
- UCD takes the needs, wants, and limitations of the **actual end users** into account **during each phase** of the design process
 - User-centered design issues are discovered during the early stages
- Benefits: systems easier to learn, with faster performance, with less human errors, encourage users to discover advanced features, and avoids “building the wrong system”
- Issues: how to find users? How many? How motivated? How to speak their language? How to extract user needs, business needs, organizational implications?

Participatory Design

- One step further than UCD, users are directly involved in the collaborative design of the things and applications they use
- Engage a group of users
 - Discussions
 - Creating scenarios, sketches, dramatizations
 - Creating and testing lo-fi prototypes
 - Continuous meetings, flexible management
 - Highly reliant on the skills of the group moderators/leaders (keep involved, filter ideas, reward participation, work around resistances, ...)
 - More effective with more mature and prepared user populations (less with kids, elderly, disabled, ...)



Agile Interaction Design

- Borrows ideas from Agile development in software engineering
- Key: evolutionary development
 - System is built incrementally in rapid release cycles
 - Rapid prototyping techniques (for hardware, software and physical objects)
- Focus on low-cost many-iterations prototypes
- Requires fast usability inspection (extreme usability, XU)
- Makers' culture (only if it involves users!)

Human-centered Design Process

A process for designing interactive systems with a focus on usability

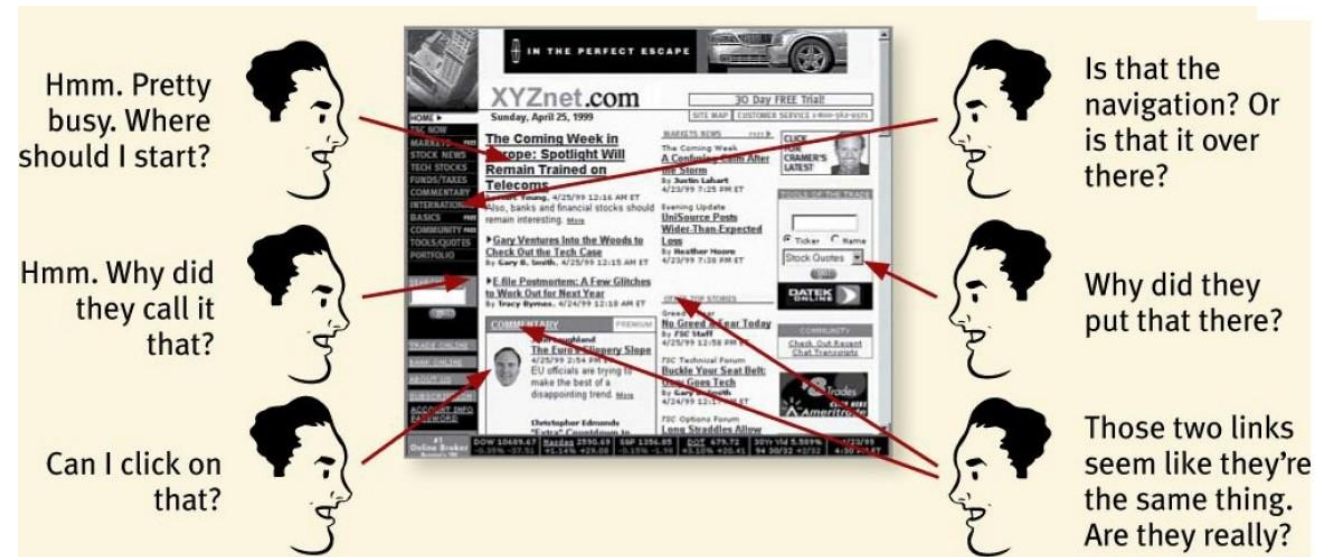
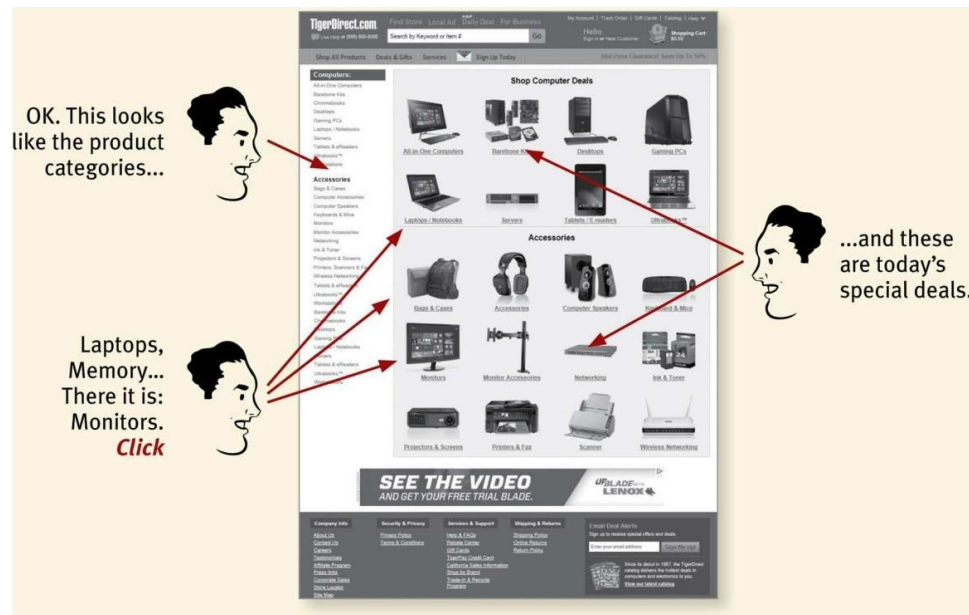
Usability (ISO standard definition)

- **Usability:** “extent to which a system, product or service can be used by specified users to achieve specified goals with *effectiveness, efficiency* and *satisfaction* in a specified context of use”
 - Note 1: The “specified” users, goals and context of use refer to the particular combination of users, goals and context of use for which usability is being considered
 - Note 2: The word “usability” is also used as a qualifier to refer to the design knowledge, competencies, activities and design attributes that contribute to usability, such as usability expertise, usability engineering, usability testing, etc.

Usability

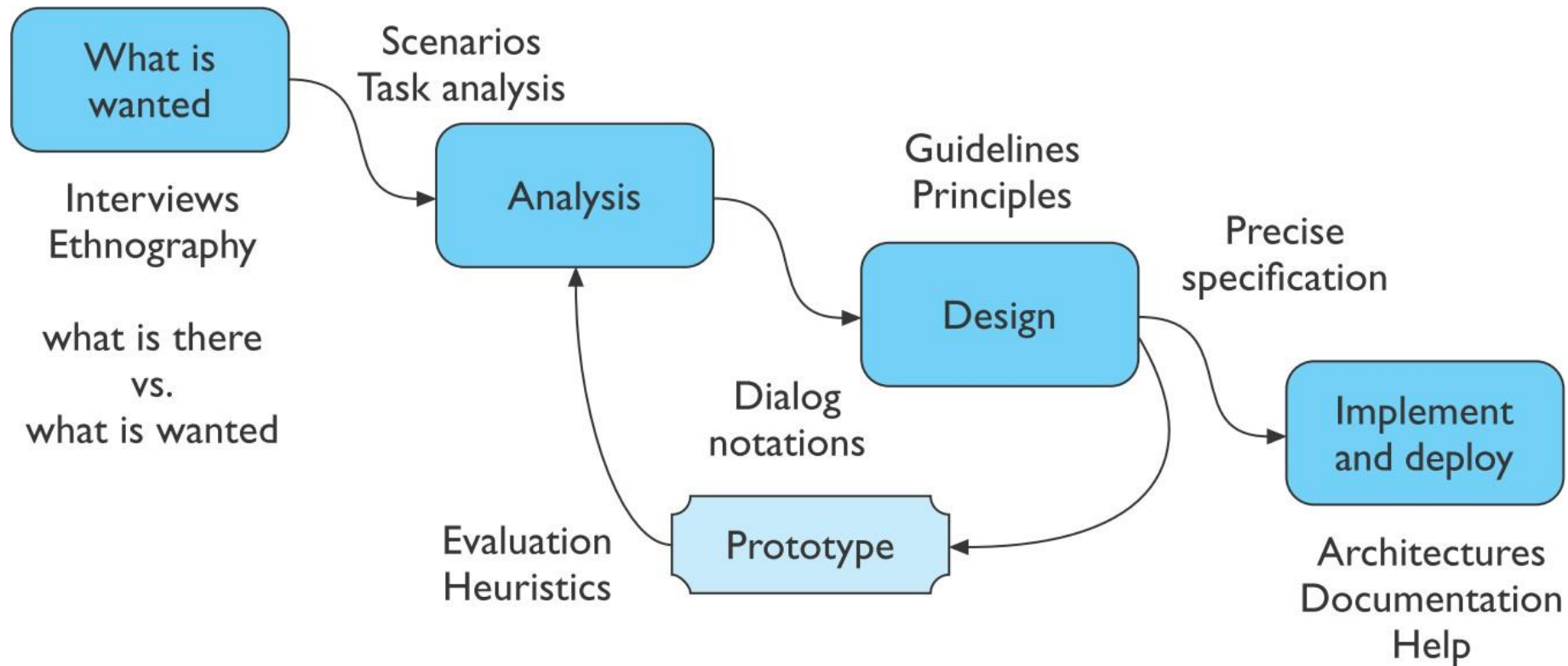
- **Usability:** how well users can use the system's functionality
- Dimensions of usability:
 - **Usefulness:** does it do something people want?
 - **Learnability:** is it easy to learn?
 - **Memorability:** once learned, is it easy to remember?
 - **Effectiveness:** does it allow reaching the goal?
 - **Efficiency:** once learned, is it fast to use?
 - **Visibility:** is the state of the system visible?
 - **Errors:** are errors few and recoverable?
 - **Satisfaction:** is it enjoyable to use?

Usability: Don't Make Me Think



Human-Centered Design Process

(simplified and generic)



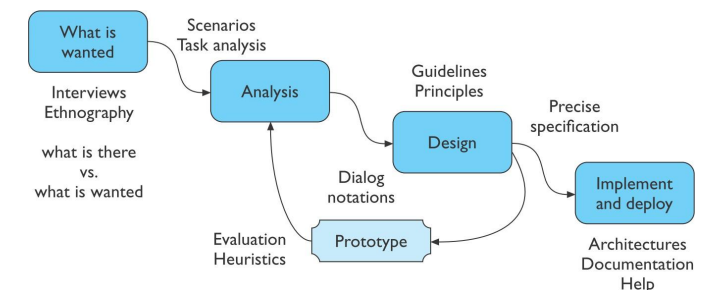
Human-Centered Design Process – The Main Steps (1)

- **Needfinding – what is wanted**

- What exactly is needed? How are people currently accomplishing the goal?
- User observation, interviews, ...

- **Analysis**

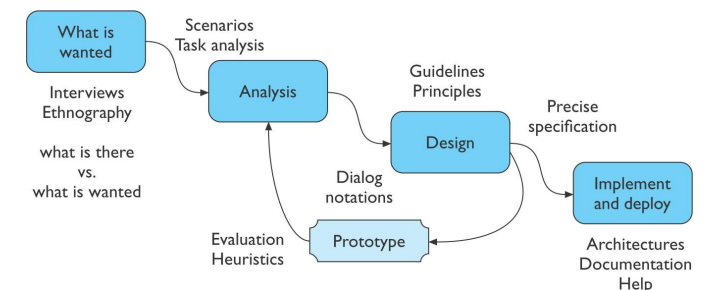
- Formalize and structure the needs
- Create interaction scenarios, stories, tasks
- Compare current situation with expected new situation



Human-Centered Design Process – The Main Steps (2)

■ Design

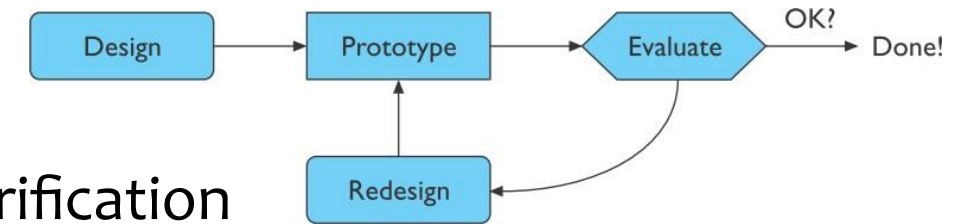
- The main choices to shape the system
- Rules, guidelines, design principles
- Considering different types of users
- Modeling and describing interaction
- Visual layout
- Consider all inputs from cognitive models, communications theories, organization issues



Human-Centered Design Process – The Main Steps (3)

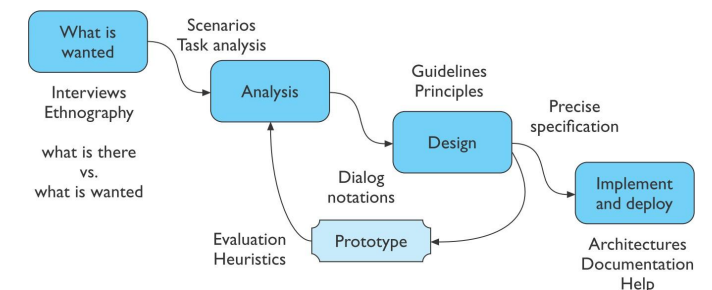
■ Iteration and prototyping

- Design must be supported by intermediate verification
- Evaluate the design in its partial forms:
 - Prototypes
 - Evaluation metrics
- Involving users

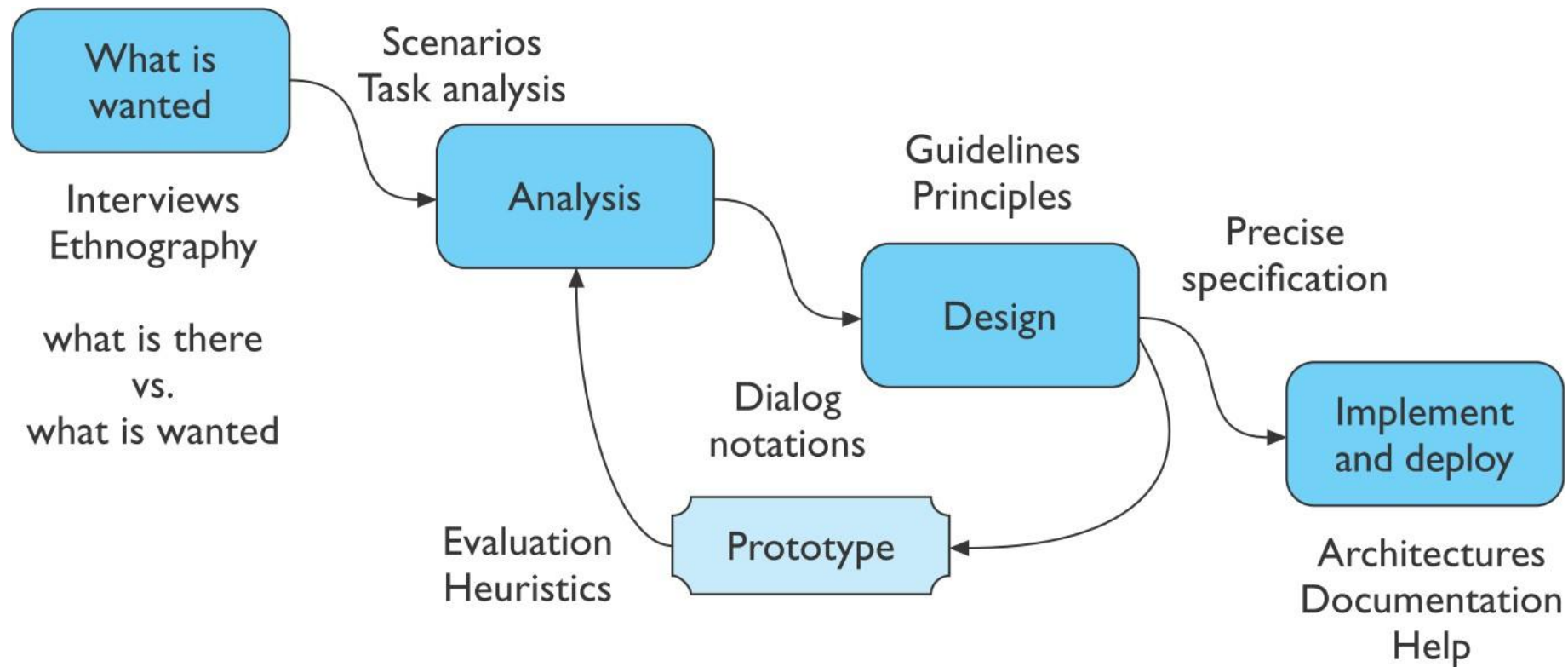


■ Implementation and deployment

- Hardware and software implementation
- Documentation



Human-Centered Design Process in this course



1 - Needfinding

2 - Prototyping (Storyboarding)

3 - Testing

Attribution

- the content of these slides is authored by Professor Luigi De Russis for the HCI module at Politecnico di Torino <https://github.com/polito-hci-2022>
- link to the license: <https://creativecommons.org/licenses/by-nc-sa/4.0/>
- changes were made by Professor Maurizio Mancini

License

- These slides are distributed under a Creative Commons license “**Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0)**”
- **You are free to:**
 - **Share** — copy and redistribute the material in any medium or format
 - **Adapt** — remix, transform, and build upon the material
 - The licensor cannot revoke these freedoms as long as you follow the license terms.
- **Under the following terms:**
 - **Attribution** — You must give [appropriate credit](#), provide a link to the license, and [indicate if changes were made](#). You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use.
 - **NonCommercial** — You may not use the material for [commercial purposes](#).
 - **ShareAlike** — If you remix, transform, or build upon the material, you must distribute your contributions under the [same license](#) as the original.
 - **No additional restrictions** — You may not apply legal terms or [technological measures](#) that legally restrict others from doing anything the license permits.
- <https://creativecommons.org/licenses/by-nc-sa/4.0/>

